

EVANTA

Event Planning & Management Platform

Project Documentation

Plan beautifully. Celebrate completely.

Team Project

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1. Project Overview

EVANTA is an event planning platform designed to help users organize and manage events from one place. The platform brings event details, guests, budget, tasks, vendors, invitations, and event planning into a single system.

The main idea is to reduce the need for scattered tools such as WhatsApp, notes, spreadsheets, and separate services. Each event has its own connected data, allowing the user to work with one current event while keeping its guests, expenses, tasks, vendors, and invitations organized.

Project Scope

- User registration and login.
- Creation and management of multiple events.
- Event-specific dashboard and planning tools.
- Guest and RSVP management.
- Budget and expense tracking.
- Task management and progress tracking.
- Vendor management.
- Digital invitations with public links and QR codes.
- Smart Plan for structured event-planning suggestions.

2. Problem Statement

Organizing an event usually requires managing several areas at the same time. Guest lists, expenses, deadlines, vendors, invitations, and RSVP responses are often stored in different tools.

Challenges Identified

- Information becomes scattered across multiple applications.
- It is difficult to maintain one clear overview of the event.
- Tasks and deadlines can be missed.
- Budget and expense information can become difficult to track.
- Guest responses may be stored separately from the event details.
- Invitation sharing and RSVP collection may require separate services.

Project Motivation

EVANTA was developed to centralize these activities so that the user can manage the event through a single platform and quickly understand the current state of the event.

3. Proposed Solution

EVANTA provides a centralized event-management workspace. After creating an event, the user can access a dashboard and dedicated sections for guests, budget, tasks, vendors, and invitations.

The platform connects event-specific records using the event ID. This keeps information belonging to different events separated while allowing the user to switch between multiple events.

Core Value

One Event → One Platform → Everything Organized

Expected Benefits

- Centralized event information.
- Faster access to important event statistics.
- Clearer tracking of guests, expenses, tasks, and vendors.
- Digital invitation and RSVP workflow.
- A structured starting point for event planning.

4. Target Users and Supported Events

Target Users

- People organizing their personal events.
- Wedding and engagement planners.
- Birthday event organizers.
- Graduation event organizers.
- Baby shower organizers.

The project presentation also identifies professional event organizers and companies as possible future users.

Currently Supported Event Types

- Birthday
- Engagement
- Wedding
- Graduation
- Baby Shower

5. User Journey and System Workflow

1. Register / Login
2. Create Event
3. Select Event Type
4. Use Smart Plan
5. Open Event Dashboard
6. Manage Guests
7. Manage Budget
8. Manage Tasks
9. Manage Vendors
10. Create Digital Invitation
11. Share Link / QR Code
12. Receive RSVP Responses
13. Monitor Everything from Dashboard

The current-event concept is central to this workflow. The selected event is stored on the frontend and used when pages request or update event-specific information.

6. Main Features

Feature	Description
Events	Create and manage multiple events and keep event-specific records connected.
Dashboard	Provides a high-level overview of event details, guests, budget, tasks, vendors, and invitations.
Guests	Add, edit, delete, search, and monitor guest RSVP status.
Budget	Add, edit, delete, categorize, and review event expenses.
Tasks	Create event-related tasks and track their progress and priority.
Vendors	Store service-provider information such as name, type, phone, price, date, status, and notes.
Invitations	Upload an invitation design, create a public link, generate a QR code, and share the invitation.
RSVP	Guests can respond without creating an account using their phone number and RSVP status.

7. Smart Plan

The Smart Plan helps the user start planning an event by selecting suitable planning options. The user can review suggested options and choose what they want to use.

Planning Inputs

- Event type
- Date
- Number of guests
- Location
- Venue
- Decoration
- Food and drinks
- Photography
- Entertainment
- Invitations
- Cake and desserts
- Flowers and details
- Budget

After selections are made, the user can review estimated cost information and determine whether the choices fit within the selected budget.

Implementation note: in the current version, Smart Plan is a planning and recommendation logic implemented in the frontend. It is not an independent AI model.

8. Event Management Screens

Dashboard

The dashboard gives the user a quick overview of the active event. The implementation shown in the supplied project screenshot includes event details, countdown information, quick statistics, RSVP snapshot, and budget overview.

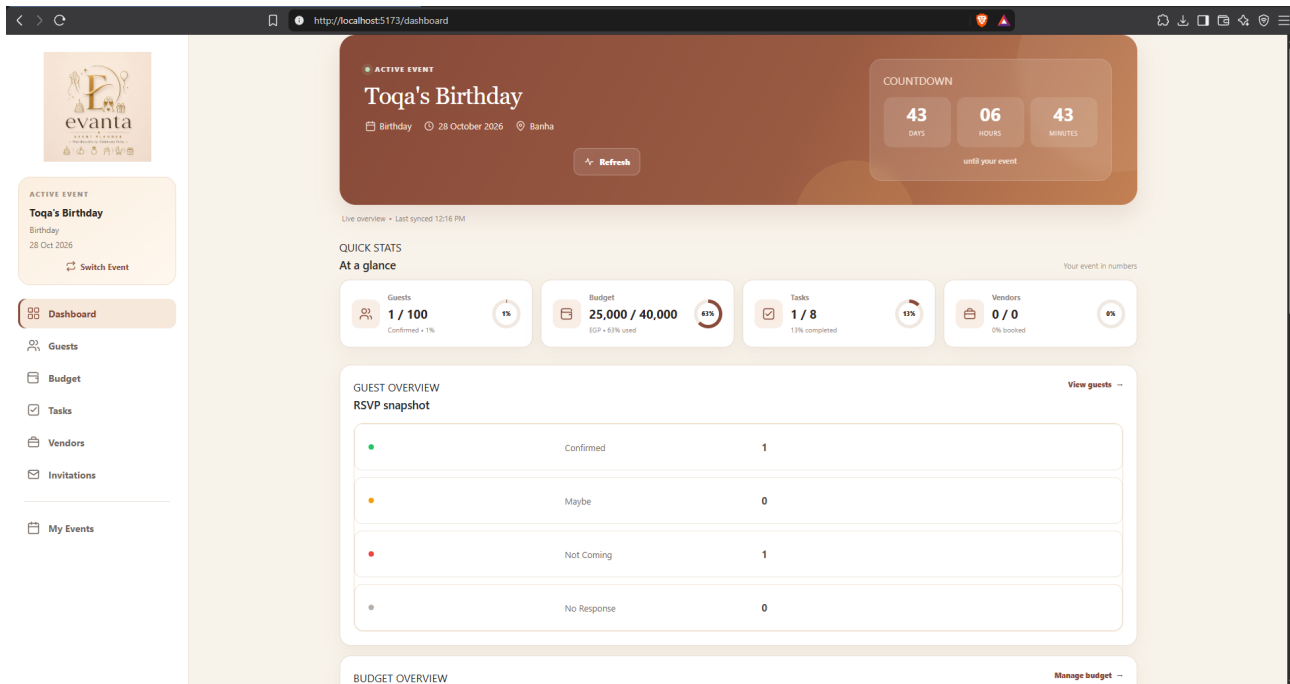


Figure 1. EVANTA event dashboard showing the active event and quick statistics.

8. Event Management Screens — Guests

The Guests page provides event-specific guest management. It shows total guests and RSVP categories, and supports searching and filtering the guest list.

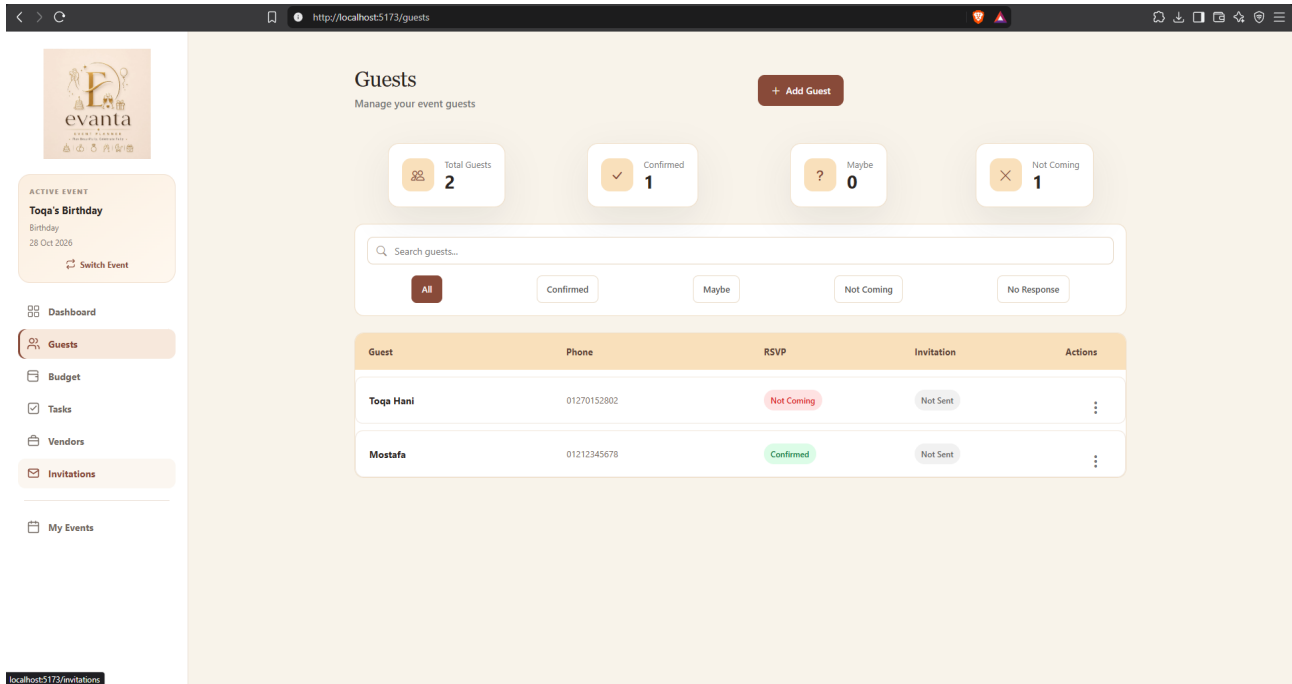


Figure 2. EVANTA Guests page with RSVP status and guest actions.

The displayed implementation includes Confirmed, Maybe, Not Coming, and No Response states.

8. Event Management Screens — Tasks

The Tasks page organizes event preparation into manageable steps. Tasks can be viewed according to status and priority, and the interface shows due dates, priority, and completion status.

The screenshot displays the EVANTA Tasks page. On the left is a sidebar with the EVANTA logo and navigation links: Dashboard, Guests, Budget, Tasks (selected), Vendors, Invitations, and My Events. The main content area is titled 'EVENT PLANNING Tasks' with a subtitle 'Keep track of everything that needs to be done.' and an '+ Add Task' button. Below this are four summary cards: Total Tasks (8), Done (1), In Progress (0), and Not Started (7). The 'Task List' section follows, with a sub-header 'Manage your event tasks and track their progress.' and filter buttons for All, Not Started, In Progress, Done, and High Priority. The task list contains seven items, each with a status icon, title, due date, priority, and completion status.

Task	Due Date	Priority	Status
Book the venue Toqa	2026-05-30	High	Not Started
Arrange the decoration	2026-05-30	Medium	Done
Prepare Drinks	2026-05-30	Medium	Not Started
Book Videographer	2026-05-30	High	Not Started
Book Games	2026-05-30	Medium	Not Started
Book Live Music	2026-05-30	Medium	Not Started
Prepare Digital Invitation	2026-05-30	Medium	Not Started

Figure 3. EVANTA Tasks page showing task progress, due dates, and priorities.

9. Digital Invitations and RSVP

EVANTA includes a digital invitation workflow. The user can upload an invitation image, generate a public invitation link, generate a QR code, and share the invitation with guests.

Invitation Workflow

- Upload the invitation design.
- Generate a public invitation link and QR code.
- Share the invitation with guests.
- Guest opens the public invitation.
- Guest submits an RSVP response.

RSVP

Guests do not need an EVANTA account to respond. They can enter their phone number and select an RSVP status.

- Confirmed
- Maybe
- Not Coming

10. Technical Architecture

EVANTA uses a separated frontend/backend architecture. The frontend communicates with the backend through HTTP requests using Axios and REST API endpoints.

React → Axios → REST API → Node.js + Express → Controllers → Mongoose → MongoDB

Backend Organization

- Routes define API paths.
- Controllers contain request-handling logic.
- Models represent database structures.
- Middleware supports protected/authenticated requests.
- Configuration handles database connection and environment-based settings.

The separation between routes, controllers, and models keeps the backend organized and makes it easier to maintain and extend.

11. Authentication and Security

EVANTA uses authentication to protect user-specific functionality and event data.

Registration

- User provides email and password.
- Password validation is performed.
- Password is hashed before storage.
- User data is stored in MongoDB.

Login

- User provides email and password.
- Credentials are validated.
- A JWT is created after successful authentication.
- Authenticated requests use the JWT.

Security Practices Used

- JWT-based authentication for protected routes.
- Password hashing using bcrypt.
- Environment variables for sensitive configuration.
- CORS configuration on the backend.
- The .env file should not be committed to GitHub.

12. Backend API Overview

Module	Endpoint	Purpose
AUTH	POST /api/auth/register	Create an account
AUTH	POST /api/auth/login	Authenticate a user
AUTH	GET /api/auth/profile	Retrieve authenticated profile
EVENTS	POST /api/events	Create an event
EVENTS	GET /api/events	Retrieve events
EVENTS	PUT /api/events/:eventId	Update an event
GUESTS	GET /api/events/:eventId/guests	Retrieve event guests
GUESTS	POST /api/events/:eventId/guests	Add a guest
GUESTS	PATCH .../guests/status	Update RSVP status
TASKS	GET /api/events/:eventId/tasks	Retrieve event tasks
TASKS	POST /api/events/:eventId/tasks	Create a task
TASKS	PUT /api/tasks/:taskId	Update a task
BUDGET	GET /api/budget/:eventId	Retrieve event expenses
BUDGET	POST /api/budget/:eventId	Create an expense
BUDGET	PUT /api/budget/:expenseId	Update an expense
INVITATIONS	GET /api/invitations/:eventId	Retrieve event invitation
INVITATIONS	POST /api/invitations/:eventId	Create/update invitation
INVITATIONS	DELETE .../invitations/:eventId	Delete invitation
VENDORS	GET /api/vendors/:eventId	Retrieve event vendors
VENDORS	POST /api/vendors/:eventId	Create vendor
VENDORS	PUT /api/vendors/vendor/:id	Update vendor
DASHBOARD	GET /api/events/:eventId/dashboard	Retrieve dashboard data

The endpoint list above reflects the endpoints documented in the project presentation. Exact backend behavior depends on the implemented modules.

13. Database Design

EVANTA uses MongoDB as its database, with Mongoose providing the data modeling and database connection layer.

Event-Specific Data

Event-related records use the event ID to keep guests, expenses, tasks, vendors, and invitations connected to the correct event.

Main Data Areas

- Users and authentication data.
- Events.
- Guests and RSVP status.
- Budget expenses.
- Tasks and their status/priority.
- Vendors and service details.
- Invitations and public invitation data.

A full database diagram can be added here if the final project submission requires an ERD image. The available project materials describe the database architecture but do not provide a database diagram image.

14. Technology Stack

Layer	Technologies
Frontend	React, Vite, JavaScript, React Router, Bootstrap, React Bootstrap, React Icons, Axios, QRCode React
Backend	Node.js, Express.js, MongoDB, Mongoose, JWT, bcrypt, dotenv, CORS

Design Language

The interface uses a warm visual palette including brown (#884a39), caramel (#c38154), peach (#ffc26f), cream (#f9e0bb), beige (#f8f3ea), white (#fdfaf5), and dark brown (#2b2522).

15. Project Structure and Setup

Project Structure

```
EVANTA
├── Frontend
│   └── Evanta-Frontend
│       └── src
│           ├── components
│           ├── pages
│           ├── context
│           ├── App.jsx
│           └── package.json
└── Backend
    ├── config
    ├── controllers
    ├── models
    ├── routes
    ├── middleware
    ├── server.js
    └── package.json
```

Environment Variables

```
PORT=3000
MONGO_URI=your_mongodb_connection_string
JWT_SECRET=your_jwt_secret
```

The .env file should remain local and should be included in .gitignore.

Installation

- Clone the repository.
- In Backend: run npm install, then npm start.
- In Frontend/Evanta-Frontend: run npm install, then npm run dev.
- The frontend runs with Vite and the backend runs as the Express server.

16. Team Contributions

Team Member	Contribution
Mariam	Home page, Login/Register, Smart Plan
Doaa	Create Event, My Events
Ahmed	Dashboard, Guests
Toqa	Budget, Invitations, Public Invitation
Shahd	Tasks
Mahmoud	Vendors

The project covers frontend development, backend development, database integration, API development, authentication, event management, and UI design.

17. Future Improvements

- AI-based event recommendations.
- Automatic budget suggestions.
- Vendor recommendations.
- More detailed RSVP statistics.
- Event analytics.
- Notification system.
- Cloud image storage.
- Deployment.
- Mobile application.

These improvements represent possible next steps beyond the current project scope.

18. Conclusion

EVANTA provides a centralized solution for planning and managing events. It combines event information, guests, budget, tasks, vendors, invitations, and RSVP responses in one platform.

The implemented system demonstrates a complete full-stack project structure using React on the frontend, Node.js and Express on the backend, and MongoDB with Mongoose for data storage. Authentication, event-specific data management, REST APIs, and a digital invitation workflow are included in the project.

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